Unfortunately, as much as we got better at forecasting, despite the amazing progresses in AI we have seen, even computers don’t have any psychic superpower, so we can’t predict *exactly* which elements we are going to need the most, and which one the least[17].

Spoiler alert: there are, however, some assumptions that we can use to try to make educated guesses.

One reasonable assumption is that, if an entry hasn’t been accessed for a long time, it has less value than an entry that was recently accessed: depending on the size of the cache and on the time elapsed till the entry was last accessed, we could infer that the oldest entry is somehow stale and it’s likely it won’t be needed soon or ever again.

On the other hand, depending on the context, even the opposite could be true: that entries that hasn’t been accessed for a long time, will be needed soon: in that case, though, removing the fresher items usually doesn’t work very well, and some other criteria must be found.

If we only look at the timestamp of last access, though, we discard other useful information; for example, if an entry was accessed several times, but not for the last few minutes, all newer entries, even if only accessed once, will be considered more valuable. If you think about it, if an entry is only required once, it means it has been computed, but never read from the cache, so that caching it has not yet brought any advantage. After some time, if that entry is yet not accessed, the utility of keeping it becomes questionable. So, a different approach could be assigning a value to entries based on how often they are accessed.